Regd. Office: Dr Lal PathLabs Ltd, Block-E, Sector-18, Rohini, New Delhi-110085
Web: www.lalpathlabs.com, **CIN:** L74899DL1995PLC065388

Name	**: Mr. VIPAN KUMAR GARG**		
Lab No.	**: 440602462**	**Age**	**: 67 Years**
Ref By	**: PGI CHANDIGARH**	**Gender**	**: Male**
Collected	**: 29/11/2024 10:39:00AM**	**Reported**	**: 4/12/2024 1:34:06PM**
A/c Status	**: P**	**Report Status**	**: Final**
Collected at	**: FPSC-Sector 8 Panchkula**	**Processed at**	**: LPL-CHANDIGARH LAB Plot No. 17, Ground Floor, Industrial Area, Phase-1, Backside Citroen Showroom, Chandigarh-160002.**

MC-6097

Test Report

Test Name	Results	Units	Bio. Ref. Interval
LIVER PANEL 1; LFT,SERUM (Spectrophotometry)			
AST (SGOT)	20.0	U/L	19.00 - 48.00
ALT (SGPT)	11.0	U/L	10.00 - 49.00
AST:ALT Ratio	**1.82**		<1.00
GGTP	27.0	U/L	0 - 73
Alkaline Phosphatase (ALP)	84.00	U/L	30.00 - 120.00
Bilirubin Total	0.30	mg/dL	0.20 - 1.10
Bilirubin Direct	0.10	mg/dL	<0.3
Bilirubin Indirect	0.20	mg/dL	<1.10
Total Protein	7.20	g/dL	5.70 - 8.20
Albumin	**4.82**	g/dL	3.20 - 4.60
A : G Ratio	**2.03**		0.90 - 2.00

Note

1. In an asymptomatic patient, Non alcoholic fatty liver disease (NAFLD) is the most common cause of increased AST, ALT levels. NAFLD is considered as hepatic manifestation of metabolic syndrome.
2. In most type of liver disease, ALT activity is higher than that of AST; exception may be seen in Alcoholic Hepatitis, Hepatic Cirrhosis, and Liver neoplasia. In a patient with Chronic liver disease, AST:ALT ratio>1 is highly suggestive of advanced liver fibrosis.
3. In known cases of Chronic Liver disease due to Viral Hepatitis B & C, Alcoholic liver disease or NAFLD, Enhanced liver fibrosis (ELF) test may be used to evaluate liver fibrosis.
4. In a patient with Chronic Liver disease, AFP and Des-gamma carboxyprothrombin (DCP)/PIVKA II can be used to assess risk for development of Hepatocellular Carcinoma.

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Test Report

Test Name	Results	Units	Bio. Ref. Interval
PROTEIN ELECTROPHORESIS, 24-HOUR URINE (Agarose Gel Electrophoresis)			
Total Protein	0.14	g/day	
Total Urine Volume	2400	mL/day	250.00 - 2400.00

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Dr Pooja Sharma
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Chief of Laboratory
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Dr Sandeep Kumar Arora
MD, Pathology
Chief of Laboratory
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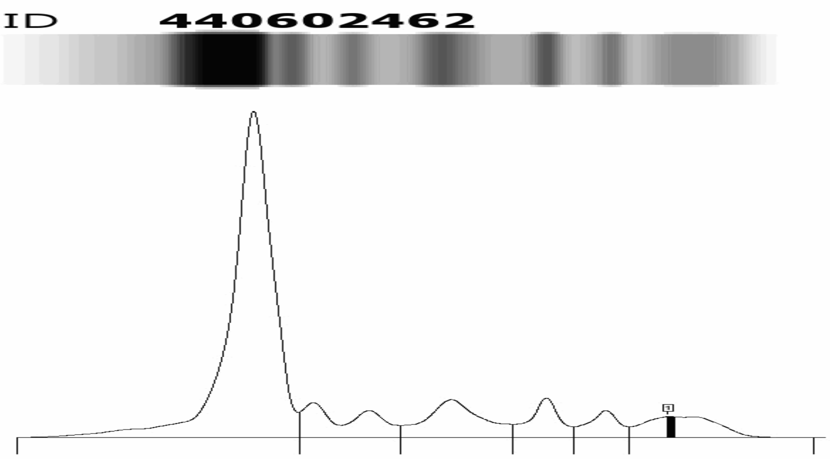
Test Report

Test Name	Results	Units	Bio. Ref. Interval
MULTIPLE MYELOMA SCREENING PANEL			
PROTEIN ELECTROPHORESIS, SERUM (Capillary Electrophoresis)			
Protein, Total	7.00	g/dL	6.40 - 8.10
Albumin	4.50	g/dL	3.60 - 5.40
Alpha 1 globulin	**0.62**	g/dL	0.20 - 0.40
Alpha 2 globulin	0.70	g/dL	0.50 - 1.00
Beta 1 globulin	**0.38**	g/dL	0.50 - 1.10
Beta 2 globulin	**0.28**	g/dL	0.30 - 0.60
Gamma globulin	**0.53**	g/dL	0.70 - 1.50
A : G Ratio	1.80		0.90 - 2.00
M Spike	? 0.1	g/dL	



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MC-2113

Test Report

Test Name	Results	Units	Bio. Ref. Interval

Interpretation

Suspicious non-discrete protein band in gamma globulin region, possibly Monoclonal gammopathy.
Advised: Immunofixation electrophoresis (IFE) to confirm the presence of monoclonal gammopathy and identify its immunological nature.

Comments

Serum Protein electrophoresis (SPE) is used to identify patients with Monoclonal gammopathies (Multiple Myeloma (MM), Waldenstrom's macroglobulinemia, AL Amyloidosis as well as premalignant conditions Monoclonal gammopathy of unkown significance(MGUS) and Smoldering Myeloma) and other serum protein abnormalities (Nephrotic syndrome, Alpha 1 antitrypsin disorder, and inflammatory processes associated with infection, liver diseases & autoimmune disorder). Monoclonal gammopathies indicate clonal expansion of plasma cells or mature B cells. Monoclonal gammopthies are present in 8% of geriatric patients and require further evaluation for monitoring & prognosis. SPE can be used for monitoring response to therapy, a decrease or increase of M spike by 0.5 g/dl is considered significant change.

If SPE alone is used as initial diagnostic screen it will be able to detect approximately 88% of all MM, 66% of AL Amyloidosis and 56% of Light Chain Deposition Disease (LCDD) patients. SPE has good sensitivity in detecting intact monoclonal protein but has limited sensitivity in detecting monoclonal free light chains (FLC). Therefore, SPE alone will miss many patients that are only producing monoclonal FLC. These misses include many Light Chain MM, all Non Secretory MM and significant numbers of AL amyloidosis and LCDD patients. Combination of FLC assay with SPE & immunofixation improves the sensitivity to 99%. Thus, Multiple Myeloma screening panel which includes SPE, IFE and FLC should be done to screen patients with high clinical suspicion for MM.

IMMUNOTYPING, SERUM .

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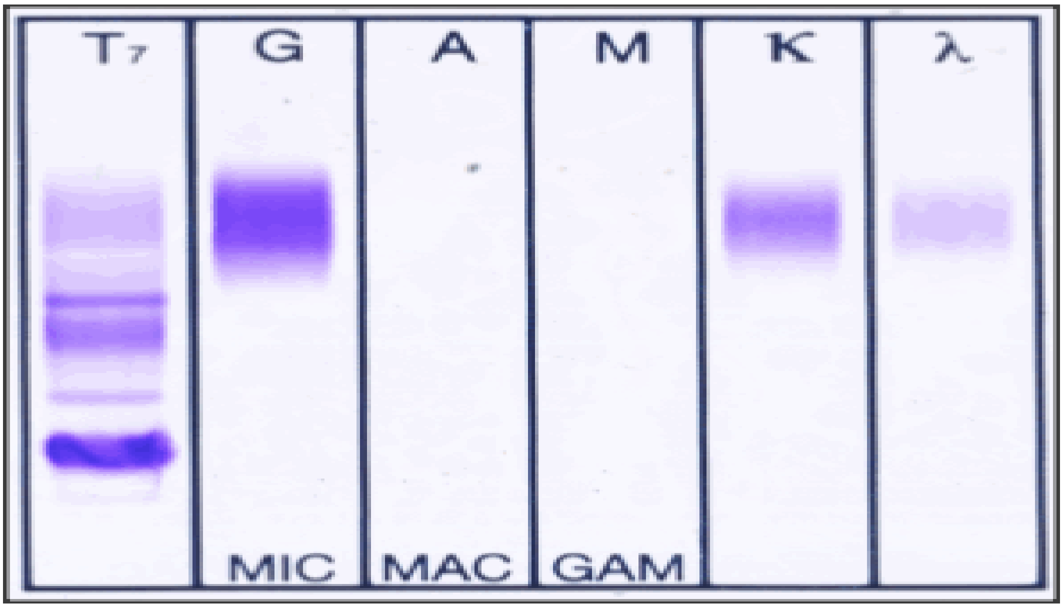
MC-2113

Test Report

Test Name	Results	Units	Bio. Ref. Interval



Interpretation:- Serum protein profile by high-resolution Agarose gel electrophoresis reveals the presence of a faint non-discrete protein band in gamma globulin region, which could be ‘M’ spike.
Immunofixation electrophoresis (IFE) identifies the very faint non-discrete protein band (?) in gamma globulin region of IgG and Kappa lane corresponding to suspicious 'M' spike.

Advised:
1. Repeat after 6-8 weeks, kindly include this report for reference.
2. Serum Kappa and Lambda free light chains (Quantitative).

Note: Immunotyping is not a quantitative assay. If a monoclonal protein is identified, a serum protein electrophoresis assay is required for quantifying the abnormality.

Comment

The assay is used for immunotyping of monoclonal proteins which identifies the monoclonal immunoglobulin heavy-chain (gamma, alpha, mu) and/or light-chain type (kappa or lambda). It is generally recommended that both serum Protein electrophoresis (SPEP) and Immunotyping be used as a screening panel because it is more sensitive than SPEP. Immunotyping is not only recommended as part of the initial screening process but also for confirmation of complete response to therapy.

Usage

Identification of monoclonal immunoglobulin heavy and light chains

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Lab No.	: 440602462	Age	: 67 Years
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MC-2113

Test Report

Test Name	Results	Units	Bio. Ref. Interval
Documentation of complete response to therapy			

Name	: Mr. VIPAN KUMAR GARG		
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Test Report

Test Name	Results	Units	Bio. Ref. Interval
IMMUNOGLOBULIN PROFILE, SERUM			
Immunogloublin IgG, Serum Immunoturbidimetry)	684.00	mg/dL	650.00 - 1600.00
Immunogloublin IgM, Serum	<21.0	mg/dL	50.00 - 300.00
Immunogloublin IgA, Serum	<33.0	mg/dL	40.00 - 350.00

Comments:

Approximately 80% of serum immunoglobulin is IgG, 6% is IgM and 13% is IgA. High levels of IgM signify acute infections whereas IgG predominates in chronic infections. IgA is the predominant immunoglobulin in body secretions like saliva, sweat and colostrums.

Polyclonal increases are seen in:

- IgG: SLE, Chronic liver diseases, Infectious diseases and Cystic fibrosis
- IgM: Viral, bacterial and parasitic infections, Liver diseases, Rheumatoid arthritis, Scleroderma, Cystic fibrosis & heroin addiction
- IgA: Chronic liver diseases, Chronic infections, Autoimmune disorders, Sarcoidosis and Wiscott-Aldrich syndrome.

Monoclonal increases are seen in IgG & IgA Myelomas and IgM increases in cases of Waldenstroms macroglobulinemia

Decreased synthesis of IgG, IgM & IgA is seen in Congenital and Acquired Immunodeficiency diseases.

Decreased levels are seen in Protein losing enteropathies, Nephrotic syndrome and skin burns.

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Test Report

Test Name	Results	Units	Bio. Ref. Interval
KAPPA / LAMBDA LIGHT CHAINS, FREE, SERUM (Immunoturbidimetry)			
Kappa, Free Light Chain	**20.31**	mg/L	3.30 - 19.40
Lambda, Free Light Chain	7.25	mg/L	5.71 - 26.30
Kappa/Lambda(FLC), Ratio	**2.801**		0.26 - 1.65

Interpretation

KAPPA	LAMBDA	KAPPA / LAMBDA RATIO	REMARKS
Normal	Normal	Normal	Normal sample. If serum electrophoretic pattern is normal, it is unlikely that patient has Monoclonal gammopathy.
Normal	Normal	Low/High	Suggestive of Monoclonal gammopathy with Bone Marrow suppression
Low	Low	Normal	Suggestive of Bone Marrow suppression
Low	Low	Low/High	Suggestive of Monoclonal gammopathy with Bone Marrow suppression
High	High	Normal	Suggestive of : • Renal Impairment • Overproduction of Polyclonal free light chain from inflammatory conditions • Biclonal gammopathy of different free light chain types
High	High	Low/High	Suggestive of Monoclonal gammopathy with Renal Impairment

Note: International Myeloma Working Group (IMWG-2014) has updated the criteria for the diagnosis of Multiple Myeloma (MM) which includes clonal bone marrow plasma cells (>or=10%) or biopsy proven Plasmacytoma, with one or more 'Myeloma defining event'. An elevated involved : uninvolved serum FLC ratio >100 (and involved FLC>or=100mg/L is now considered a 'Myeloma defining event'.

Comment

Measurement of free light chain (FLCs) aids in the diagnosis & monitoring of Multiple myeloma, Waldenstrom's macroglobulinemia, AL Amyloidosis & Light chain deposition disease. Serum concentration of FLCs is dependent upon the balance between production by plasma cells & their progenitors and renal clearance.

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Test Report

Test Name	**Results**	**Units**	**Bio. Ref. Interval**

When there is increased polyclonal immunoglobulin production and/or renal impairment, concentration of both kappa & lambda light chains can increase by 30-40 fold but the Kappa / Lambda ratio remains unchanged. In contrast, tumor produces monoclonal excess of only 1 type of light chains, often with bone marrow suppression of alternate light chains resulting in abnormal kappa/lambda ratio.

The combination of Serum protein electrophoresis (SPE) & Immunofixation (IFE) are highly sensitive but fail to identify a few patients with Light Chain Multiple Myeloma & all patients of Non Secretory Multiple Myeloma. Combination of free light chain assay with SPE & IFE improves the sensitivity to 99%.

Uses

As per recommendations by IMWG:

- FLC,SPE and serum IFE for screening
- For diagnosis and prognosis of all patients with MGUS, Smoldering or active multiple myeloma, Solitary plasmacytoma and AL amyloidosis
- Serial measurements routinely in patients with AL amyloidosis and Multiple myeloma
- patients with oligosecretory disease
- In all patients who have achieved a complete response (CR) to assess for stringent CR

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MC-2113

Test Report

Test Name	Results	Units	Bio. Ref. Interval
PROTEIN ELECTROPHORESIS, 24-HOUR URINE (Agarose Gel Electrophoresis)			
Total Protein	0.14	g/day	0.04 - 0.15
Total Urine Volume	2400		
Protein Electrophoresis	**Interpretation:-**High resolution urine protein electrophoresis does not reveal the presence of any abnormal bands. No "M" Spike seen.		

Comments

Protein electrophoresis evaluates the major protein fractions and determines their deficiencies and excesses as seen with Macroglobulinemia, MGUS & Multiple myeloma.This assay is recommended specially in cases of hypoproteinemia / hypogammaglobulinemia where serum levels are low but urine detects abnormal proteins.

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MCI NO: 15-19066
Dr.Richa Sirohi
MD, Biochemistry
Sr. Consultant Biochemist
NRL - Dr Lal PathLabs Ltd

------------------------------End of report --------------------------------

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Test Report

Test Name	Results	Units	Bio. Ref. Interval

IMPORTANT INSTRUCTIONS

•Test results released pertain to the specimen submitted.•All test results are dependent on the quality of the sample received by the Laboratory. •Laboratory investigations are only a tool to facilitate in arriving at a diagnosis and should be clinically correlated by the Referring Physician.•Report delivery may be delayed due to unforeseen circumstances. Inconvenience is regretted.•Certain tests may require further testing at additional cost for derivation of exact value. Kindly submit request within 72 hours post reporting.•Test results may show interlaboratory variations.•The Courts/Forum at Delhi shall have exclusive jurisdiction in all disputes/claims concerning the test(s) & or results of test(s).•Test results are not valid for medico legal purposes.•This is computer generated medical diagnostic report that has been validated by Authorized Medical Practitioner/Doctor.•The report does not need physical signature.

(#) Sample drawn from outside source.

If Test results are alarming or unexpected, client is advised to contact the Customer Care immediately for possible remedial action.

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National Reference lab, Delhi, a CAP (7171001) Accredited, ISO 9001:2015 (FS60411) & ISO 27001:2013 (616691) Certified laboratory.